Ishani Deshpande

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OBJECTIVE

Computer Engineering student at Georgia Tech with experience testing Raspberry Pi embedded systems, measuring electrical performance, and developing computer vision models for retinal imaging. Seeking a Summer 2027 Electrical & Computer Engineering Internship to contribute to embedded software development, laboratory system testing, and eye-care technology.

EDUCATION

Georgia Institute of Technology | Atlanta, GA
Bachelor of Science in Computer Engineering
Expected May 2029

TECHNICAL SKILLS

- **Programming:** Python, Java, C/C++, Verilog, SystemVerilog, OOP
- **Embedded Systems:** Raspberry Pi, Embedded Systems, Linux/UNIX, Digital Logic
- **Hardware & Testing:** Analog Discovery 2, Electrical Measurement, Shunt-Based Sensing, Hardware Validation, System Testing, Performance Characterization, Digital Logic, Computer Architecture (CPU/GPU)
- **AI/ML:** PyTorch, NumPy, Matplotlib, Computer Vision, Machine Learning, OpenAI API
- **Software/Tools:** Git, GitHub, VS Code, Microsoft Excel
- **Software Development:** FastAPI, REST APIs, React

RESEARCH EXPERIENCE

Worcester Polytechnic Institute | Worcester, MA
Embedded Systems Research Intern
May 2026 – August 2026

- Evaluated embedded AI workloads on Raspberry Pi 5 for computer vision applications, measuring latency and system performance across workloads.
- Measured voltage, current, and power using Analog Discovery 2 and shunt-based sensing to characterize system performance and energy efficiency.
- Debugged Linux-based camera pipelines and embedded software to improve reliability and reproducibility of performance testing.

UMass Chan Medical School | Worcester, MA
Macular Degeneration Data Analysis Intern
June 2024 – August 2024

- Analyzed 60+ retinal microscopy images and experimental datasets to quantify treatment effects and identify disease-related trends
- Prepared retinal tissue and performed microscopy, PCR, and biological staining to support experimental data collection

PROJECTS

Deep Learning for Age-Related Macular Degeneration | Python, PyTorch, Computer Vision, Machine Learning

- Developed and evaluated a deep learning model to detect age-related macular degeneration from retinal images
- Preprocessed and classified 1,000+ retinal scans using computer vision and machine learning techniques
- Achieved 97% classification accuracy on 1,000+ retinal scans; evaluated model performance using precision, recall, and F1-score

Laboratory Workflow Simulator | C++

- Built a CSV-driven simulator for laboratory sample transfers, implementing input validation and per-well volume tracking to reject invalid instructions and prevent overfilling.
- Generated transfer execution logs and developed tests covering valid workflows, malformed inputs, and well-capacity limits.